

2 Flowering plants produce seeds that can be stored in a seed bank. This is done to conserve the biodiversity of plants.

(a) (i) Describe how seeds are treated and then stored in a seed bank.

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(ii) State what is meant by the term **biodiversity**.

(2)

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- (b) The longer seeds are stored in seed banks, the less likely they are to germinate.
This is due to ageing.

Ageing of seeds can affect the growth of roots when the seeds germinate.

Determining the mitotic index of roots is one way to study growth of roots.

- (i) Describe how the mitotic index of roots can be determined.

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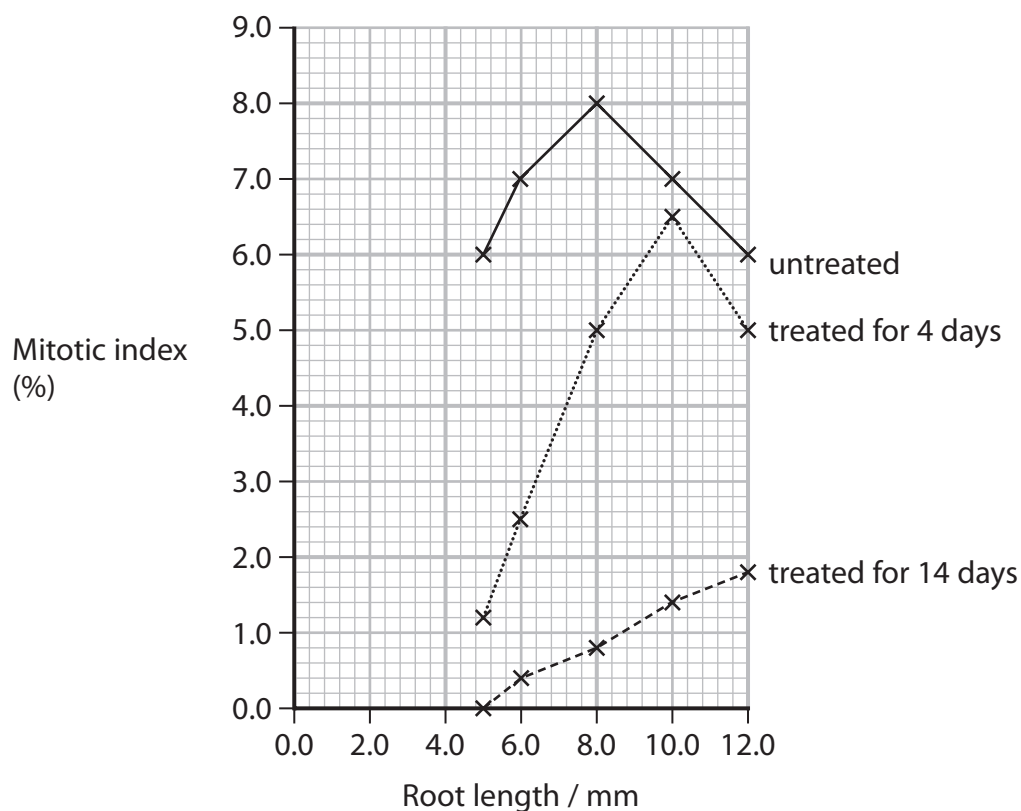
(ii) Seeds can be treated to artificially age them.

In an investigation, untreated and treated samples of seeds were germinated and the roots allowed to grow.

Roots of different lengths from the untreated seeds, seeds treated for 4 days and seeds treated for 14 days were selected.

The mitotic index of these roots was determined.

The graph shows the results of this investigation.



There were 14 cells undergoing mitosis in 8 mm roots from seeds treated for 4 days.

Calculate the total number of cells that were counted when the mitotic index was determined.

(3)

Total cells counted



- (iii) Draw a table to show the data on the effect of root length on mitotic index in seeds treated for 14 days.

(3)

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P 6 5 8 0 9 A 0 9 1 6

(iv) Identify the conclusions that can be drawn from these data.

(3)

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(v) The graph does not show the standard deviations of the means for mitotic index.

Describe how the investigation could be modified to allow standard deviations to be obtained.

(2)

(Total for Question 2 = 20 marks)

